V.Imp ☆☆☆

Regularization part 1| bias variance trade-off

Bias Variance Trade-off

- → Bias
- → Variance
- → Overfitting.
- → Underfitting.
- → Trade-off
- → Bias variance decomposition.

The Hidden Truth :

n=100

cgpa	iq	lpa
8	80	8
9	90	9
—	—	—
—	—	—

→ model → predict → cgpa|iq → lpa.

X Y

x_1 x_2 x_n

Our job is to find formula b/w x, y

$y = f(x)$

When we put value of cgpa|iq → predicts lpa.

Problem is we have to make formula for all student with n=100 given data.

(population) (sample)

It's difficult task.

$$y = f(x) + \underset{\uparrow \text{ irreducible error.}}{\overset{\text{noise in data.}}{\text{error}}}$$

Now, we try to get best estimate with f(x).
But we have to predict for population. Thus we are able to find f'(x)
f'(x) is a close estimate of f(x).
∴ The predicted values are $\hat{y}$ not y.

$$y - \hat{y} \qquad \underbrace{f(x) - f'(x)}_{\text{reducible error.}}$$

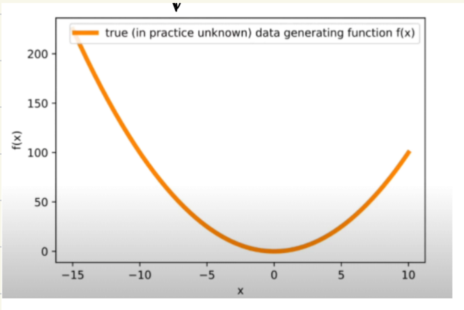
: If we get more n, might be possible to reduce the error.

Bias Variance Trade-off. :

sample ⟶ model ⟶ predict.

What we will do for now:
population ⟶ sample.

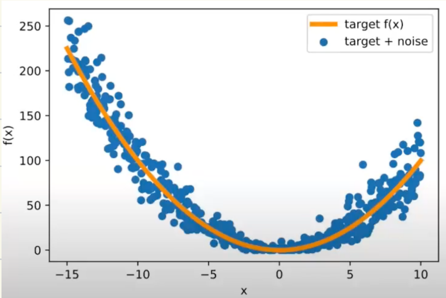
Population fn. $y = f(x) = x^2$ Range: [-15, 10].



Population fn. graph.

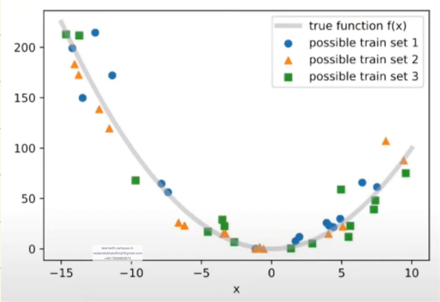
In real world we don't have population data.

We added the noise to the population fxn and generated population data.



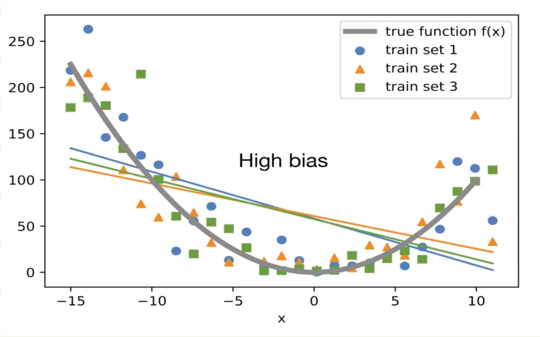
← $y = f(x) + \text{error}$.

Now we draw 3 random samples. : • → sample 1   ▲ → Sample 2   ■ → Sample 3



Given the sample data to 3 different set of people to fit ML model.

All 3 students used LR model.



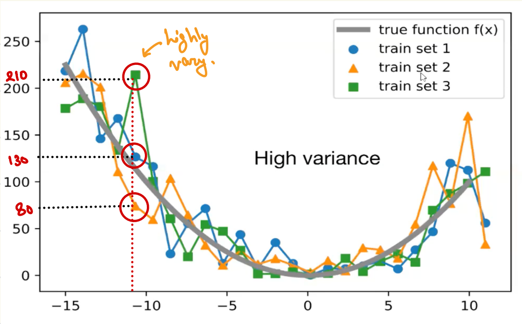
← High Bias Low Variance.

Linear models.

Bias → The inability of ML model to fit training data.

Variance → How much does the ML model's prediction changes when the training data is changed.

Now the students fit polynomial regression, (degree = 2).



Low bias, high variance.

The model is best fitted with one set of sample data. Once a new sample is drawn the result is not good.

Overfitting → When the model predicts well for training data but does not give good results for new sample.

High variance is closely related to overfitting.

Underfitting → When the model does not predict well for training data.

High bias is closely related to underfitting.

The "trade-off" in bias-variance trade off refers to the fact that minimizing bias will usually increase variance and vice-versa.

As a ML Engineer we have to find a ML model that does not under/over fit.

Some Questions :

1. How would you define bias and variance mathematically?
2. How is bias and variance related to overfitting and underfitting mathematically?
3. Why is there a trade-off b/w bias and variance mathematically?

Expected Value and Variance :

Expected value represents the average outcome of a random variable over a large number of trials or experiments.

eg.) X = Rolling a die.

n = 1 Lakh times.

$$E[X] = x_1 P(x_1) + x_2 P(x_2) + \dots + x_n P(x_n)$$

$\uparrow$ Discrete Random Var.

$$E[X] = 1 \cdot P(1) + 2 \cdot P(2) + 3 \cdot P(3) + 4 \cdot P(4) + 5 \cdot P(5) + 6\, P(6)$$

$$E[X] = 1 \cdot \left(\tfrac{1}{6}\right) + 2\left(\tfrac{1}{6}\right) + 3\left(\tfrac{1}{6}\right) + 4\left(\tfrac{1}{6}\right) + 5\left(\tfrac{1}{6}\right) + 6\left(\tfrac{1}{6}\right)$$

$$E[X] = 1 + 2 + 3 + 4 + 5 + 6 \cdot \left(\tfrac{1}{6}\right) = \frac{21}{6} = 3.5$$

$$E[X] = 3.5$$

$\uparrow$ Expected value.

$\boxed{\text{Expected value} \approxeq \text{Population mean.}}$

Discrete Random Variable : $E[x] = \sum_{i=1}^{n} x_i P(x_i)$

Continuous Random Variable : $E[x] = \int x_i P(x_i) dx.$

Var(x) : Variance of Population :

$$Var(x) = E[x^2] - (E[x])^2$$

$\Sigma \left(\frac{\cdot}{n}\right) = E[x]$

population

$E[x+Y] = E[x] + E[Y]$

$E[x \cdot Y] = E[x] E[Y]$

x, y are independent

$E[const] = const$

Derivation :

sample mean.

$$Var = \underbrace{\sum_{i=1}^{n} \frac{\left(x_i - \bar{x}\right)^2}{n}}_{avg.}$$

$$var(x) = E\left[(x - E[x])^2\right]$$

$$var(x) = E\left[(x - E[x])^2\right]$$

$$= E\left[x^2 + (E[x])^2 - 2x\, E[x]\right]$$

Rule: $E[x+Y] = E[x] + E[Y]$

const Random const.

var

$$= E[x^2] + E\left[(E[x])^2\right] - E\left[2x\, E[x]\right]$$

Rule: $E[x\, Y] = E[x] \cdot E[Y]$

$$= E[x^2] + E\left[(E[x])^2\right] - E[2] \cdot E[x] \cdot E[E[x]]$$

Rule: $E[const] = const.$

$$= E[x^2] + E\left[(E[x])^2\right] - 2 \cdot E[x]\, E[x]$$

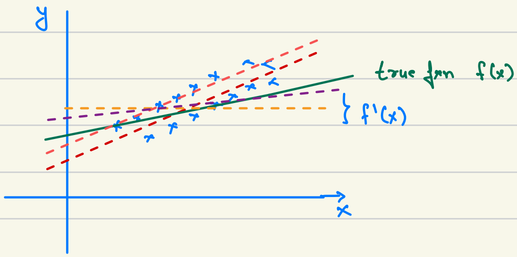
$$= E[x^2] + E\left[(\underbrace{E[x]}_{const.})^2\right] - 2(E[x])^2$$

$$= E[x^2] + (E[x])^2 - 2(E[x])^2$$

For both discrete & continuous : $\boxed{var(x) = E[x^2] - (E[x])^2}$

$$var(x) = E\left[(x - E[x])^2\right]$$

1. How would you define bias and variance mathematically?



If we take 100 $f'(x)$ then, $E[f'(x)]$

$$\boxed{Bias = E[f'(x)] - f(x)}$$

If our $E[f'(x)] = 0$ i.e we get exact $f(x)$, then $Bias = 0$

$Bias = 0$: Unbiased predictor.

2. How is bias and variance related to overfitting and underfitting mathematically?

$$\boxed{var\ f'(x) = E\left[\left(f'(x) - E[f'(x)]\right)^2\right]}$$

$$var(x) = E\left[\left(x - E[x]\right)^2\right]$$

If var $f'(x)$ is high, test data will vary a lot. Leads to Overfitting.

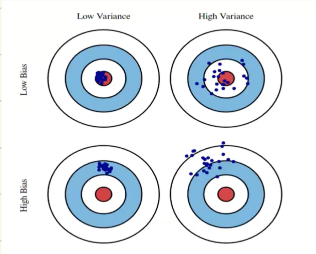


Fig. 1 Graphical illustration of bias and variance.

Regularization part 2 | What is regularization?

$$Bias(f'(x)) = E[f'(x)] - f'(x)$$

$$E[\hat{\theta}] - \hat{\theta}$$

$$Var(f'(x)) = E[(f'(x) - E[f'(x)])^2]$$

Bias - Variance Decomposition :

Bias-Variance decomposition is a way of analysing a learning algo.'s expected generalizing error with respect to a particular problem by expressing it as the sum of three very different quantities : bias, variance, and irreducible error.

Loss = bias + variance + irreducible error.

$$Loss = \underbrace{bias^2 + var}_{\text{reducible error}} + \underbrace{var(\epsilon)}_{\text{irreducible error.}}$$

(Noise)

Derivation :

cgpa	iq	lpa	prediction
—	-	8	9
—	-	8	8.1
—	-	7	6.5
—	—	9	10.1

$$\sum_{i=1}^{n} (y - \hat{y})^2 \leftarrow \text{minimize error.}$$

$$y = \beta_0 + \beta_1 \, cgpa + \beta_2 \, iq$$

[error = irreducible + reducible]

Let, irreducible error. mean $= 0$

irreducible error . var $= \sigma^2$

MSE (Mean Sq. Error) $= \dfrac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{n}$

$= E[(Y-\hat{Y})^2]$

true $y = f(x) + \varepsilon$, $\quad f(x) \to \theta$

$\hat{y} = f'(x)$, $\quad f'(x) \to \hat{\theta}$

$= E[(\underbrace{\theta + \overset{b}{\varepsilon} - \hat{\theta}}_{a})^2]$

$(a+b)^2 = a^2 + b^2 + 2ab$

$= E[\underbrace{(\theta-\hat{\theta})^2}_{a^2} + \underbrace{\varepsilon^2}_{b^2} + \underbrace{2\varepsilon(\theta-\hat{\theta})}_{2ab}]$

$E[X+Y] = E[X] + E[Y]$

$= E[(\theta-\hat{\theta})^2] + E[\varepsilon^2] + E[2\varepsilon(\theta-\hat{\theta})]$

breaking it down.

$E[XY] = E[X]\,E[Y]$

$= E[(\theta-\hat{\theta})^2] + E[\varepsilon^2] + E[2]\,\underbrace{E[\varepsilon]}_{0}\,E[\theta-\hat{\theta}]$

0, as mean is 0.

$= E[(\theta-\hat{\theta})^2] + E[\varepsilon^2]$

$E[\varepsilon^2] = ?$

$var(\varepsilon) = \sigma^2 = E[(\varepsilon - E[\varepsilon])^2]$

but $E[\varepsilon] = 0$

$\therefore \quad var(\varepsilon) = \sigma^2 = E[\varepsilon^2]$

$$MSE = \underbrace{E[(\theta - \hat{\theta})^2]}_{\text{solving this}} + var(\varepsilon)$$

$$E[(\theta-\hat{\theta})^2] = E[(\overbrace{\theta - E[\hat{\theta}]}^{a} + \overbrace{E[\hat{\theta}] - \hat{\theta}}^{b})^2]$$

$$= E\left[(\theta - E[\hat{\theta}])^2 + (E[\hat{\theta}]-\hat{\theta})^2 + 2(\theta - E[\hat{\theta}]).(E[\hat{\theta}]-\hat{\theta})\right]$$

Rules: $E[X.Y] = E[X]\,E[Y]$, $E[X+Y] = E[X] + E[Y]$

$$= E[(\theta - E[\hat{\theta}])^2] + E[(E[\hat{\theta}]-\hat{\theta})^2] + \underbrace{E[2(\theta - E[\hat{\theta}]).(E[\hat{\theta}]-\hat{\theta})]}$$

$$\underbrace{\overset{\text{const}}{E[2]}\ \overset{\text{const}}{E[(\theta - E[\hat{\theta}])]}\ E[(E[\hat{\theta}]-\hat{\theta})]}$$

$$2(\theta - E[\hat{\theta}])\{E[E[\hat{\theta}]] - E[\hat{\theta}]\}$$

$$\underbrace{\{E[\hat{\theta}] - E[\hat{\theta}]\}}_{0}$$

$$= 0$$

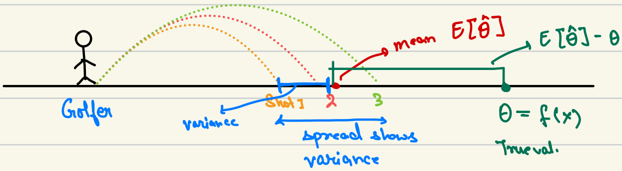
$$= \underbrace{E[(\theta - E[\hat{\theta}])^2]}_{\substack{(\theta - E[\hat{\theta}])^2 \\ (Bias)^2}} + \underbrace{E[(E[\hat{\theta}]-\hat{\theta})^2]}_{var}$$

$$\boxed{MSE = \underbrace{(Bias)^2 + var}_{\text{reducible error}} + \underset{var(\varepsilon)}{\text{irreducible}}}$$

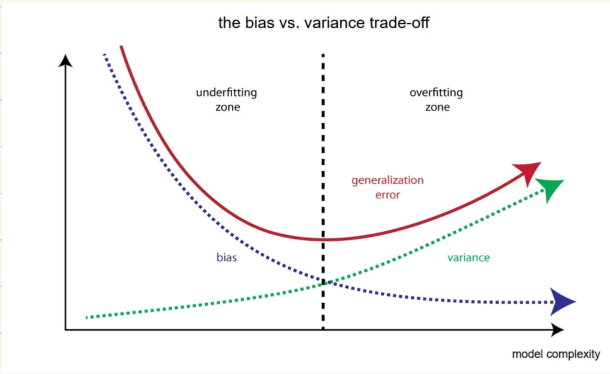
Analogy :

$$MSE = (Bias)^2 + variance + var(\varepsilon)$$

Let $var(\varepsilon) = 0$



$\hat{\theta} = f(x)$



Ways to reduce bias : Complex Models , Neural Networks.

Ways to reduce var : Regularization

What is Regularization ?

$$\text{Loss} = (\text{bias})^2 + \text{var} + \text{noise}.$$

Complex models. } But it lead to high var.

∴ We use Regularization.

Regularization is used in many ML model.
It's a universal technique.

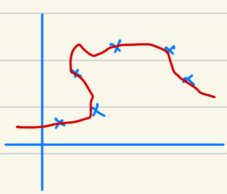
Linear Regression :

$$L = \sum_{i=1}^{n} \frac{(Y_i - \hat{Y}_i)}{n}$$

$$\Big[\underbrace{(L) + \text{extra term}}_{\text{Regularization.}}\Big]$$

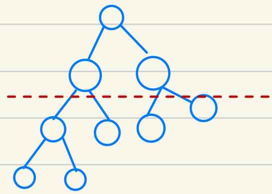
After minimization, overfitting is reduced.

Decision Tree



tree?

(tree : nested if else statement)



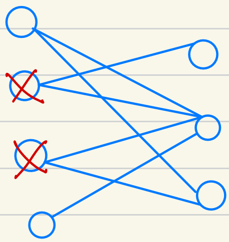
Prune the tree (cut)

Now tree is less complex hence, reduces var.

Deep Learning :

Randomly delete node

(Drop Out)



Less complex model, thus, var ↓.

When to use Regularization ?

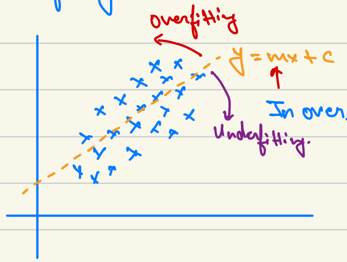
1. Preventing Overfitting.
2. High Dimensionality.
3. Multicollinearity.
4. Feature Selection.
5. Interpretability.
6. Model Performance.

Ridge regression | Part 1

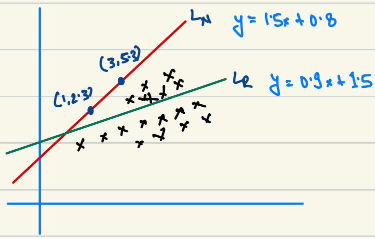
Regularization → Adding some element in ML model to reduce overfitting

Ridge Regularization also called L2

Overfitting



In overfitting m is very high.



Line LN is training data

Line LR is for test data.

LR is best fit line, but LN is the line we get after training. LN will do more mistakes on training data.

$$L = \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{n}$$

Regularizing L : $L = \sum_{i=1}^{n} \frac{(y - \hat{y}_i)^2}{n} + \lambda(m^2)$

m → slope.

λ → hyperparameter $[0, \infty)$

Loss LN	Loss LR.
$\lambda = 1$	$\lambda = 1$
$= 0 + (1.5)^2$	$= (2.3 - 1.9 - 1.5)^2 + (5.3 - 2.7 - 1.5)^2 + (0.9)^2$
$= 2.25$	$= (0.1)^2 + (1.1)^2 + (0.9)^2$
	$= 2.03$

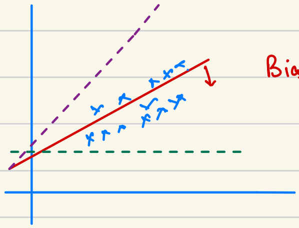
Why it's called L2 regularization ?

$$\lambda (m_1^2 + m_2^2 + m_3^2 + \dots m_n^2)$$

Since we multiply with square

$\therefore$ L^2 Norm.

Part 2 (Mathematical Intuition)



$$y = mx + b$$

L.R : $\displaystyle L = \sum_{i=1}^{n} \frac{(y_i - \hat{y}_i)^2}{n}$

$$L = \sum_{i=1}^{n} \frac{(y_i - \hat{y}_i)^2}{n} + \lambda (m^2)$$

We have to reduce m.

$$L = \sum_{i=1}^{n} (y_i - mx_i - b)^2 + \lambda m^2$$

$$\frac{\partial L}{\partial b} = 0$$

$$\frac{\partial L}{\partial m} = 0$$

$$\frac{\partial L}{\partial b} = \bar{y} - m\bar{x}$$

$$\frac{\partial L}{\partial m} = (y_i - m x_i - \bar{y} + m\bar{x})^2 + \lambda(m^2)$$

$$\frac{\partial L}{\partial m} = 2 \sum_{i=1}^{n} (y_i - m x_i - \bar{y} + m\bar{x}) \cdot (-x_i + \bar{x}) + 2\lambda m = 0$$

$$\frac{\partial L}{\partial m} = -2 \sum_{i=1}^{n} (y_i - m x_i - \bar{y} + m\bar{x}) \cdot (x_i - \bar{x}) + 2\lambda m = 0$$

$$\frac{\partial L}{\partial m} = \lambda m - \sum_{i=1}^{n} \big((y_i - \bar{y}) - m(x_i - \bar{x})\big) \cdot (x_i - \bar{x}) = 0$$

$$\frac{\partial L}{\partial m} = \lambda m - \sum_{i=1}^{n} \left[(y_i - \bar{y})(x_i - \bar{x}) - m(x_i - \bar{x})^2\right] = 0$$

$$\frac{\partial L}{\partial m} = \lambda m - \sum_{i=1}^{n} (y_i - \bar{y})(x_i - \bar{x}) + m\sum_{i=1}^{n} (x_i - \bar{x})^2 = 0$$

$$\frac{\partial L}{\partial m} \Rightarrow \lambda m + m\sum_{i=1}^{n} (x_i - \bar{x}) = \sum_{i=1}^{n} (y_i - \bar{y})(x_i - \bar{x})$$

$$\frac{\partial L}{\partial m} \Rightarrow m = \sum_{i=1}^{n} \frac{(y_i - \bar{y})(x_i - \bar{x})}{(x_i - \bar{x})^2 + \lambda}$$

For normal L·R $\quad m = \sum_{i=1}^{n} \frac{(y_i - \bar{y})(x_i - \bar{x})}{(x_i - \bar{x})^2}$

$\lambda \rightarrow$ Hyperparameter (alpha)

$\lambda = 0$ m const. ; $\lambda \rightarrow \uparrow$ $m \downarrow$; $\lambda \rightarrow \downarrow$ $m \rightarrow \uparrow$

Ridge Regression for N-Dims Data :

$w_0 \quad x_1, \; x_2, \; x_3, \; x_4 \; \cdots\cdots, \; x_n$

$\quad\quad w_1 \quad w_2 \quad\quad\quad\quad\quad\quad w_n \rightarrow$ Coefficient.

$\quad x_{1m} \; \cdots\cdots\cdots \; x_{nm}$

$$L = \sum_{i=1}^{m} (y_i - \hat{y}_i)^2$$

$$(XW - Y)^T (XW - Y)$$

$$\begin{array}{c} \text{m vals} \\ Y = \begin{bmatrix} - \\ - \\ - \\ - \end{bmatrix}_{(1\times m)} \\ \text{Output vals} \end{array} \quad W = \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_n \end{bmatrix}_{(1\times m)} \quad X = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1n} \\ 1 & x_{21} & x_{22} & \cdots & x_{2n} \\ 1 & \vdots & \vdots & \cdots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nm} \end{bmatrix}_{(n\times m)}$$

$$L = (XW - Y)^T (XW - Y) + \lambda \underline{\|W\|^2}$$

$$\rightarrow \begin{cases} \lambda w_0^2 + \lambda w_1^2 + \cdots + \lambda w_n^2 \\ \lambda (w_0^2 + w_1^2 + \cdots + w_n^2) \Rightarrow W^T W \end{cases}$$

$$L = (XW - Y)^T (XW - Y) + \lambda W^T W$$

$$L = ((XW)^T - Y^T)(XW - Y) + \lambda W^T W$$

$(a-b)^T = a^T - b^T$

$$L = (W^T X^T - Y^T)(XW - Y) + \lambda W^T W$$

$(ab)^T = b^T a$

$$L = w^T x^T x w - \boxed{w^T x^T y - y^T x w} + y^T y + \lambda w^T w$$

Same Term (Refer Multiple L.R)

$$L = w^T x^T x w - 2 w^T x^T y + y^T y + \lambda w^T w$$

Scalar Derivative $f(x) \longrightarrow \frac{df}{dx}$	Vector Derivative $f(x) \longrightarrow \frac{df}{dx}$
$bx \longrightarrow b$	$x^T B \longrightarrow B$
$bx \longrightarrow b$	$x^T b \longrightarrow b$
$x^2 \longrightarrow 2x$	$x^T x \longrightarrow 2x$
$bx^2 \longrightarrow 2bx$	$x^T B x \longrightarrow 2Bx$

$$L = w^T x^T x w - 2 w^T x^T y + y^T y + \lambda w^T w$$

$$\frac{\partial L}{\partial w} = 2 x^T x W - 2 x^T y + 0 + 2 \lambda W = 0$$

$$\frac{\partial L}{\partial w} \Rightarrow x^T x w + \lambda w = x^T y$$

$$\frac{\partial L}{\partial w} \Rightarrow w(x^T x + \lambda I) = x^T y$$

[I = Identity Matrix, as we cannot keep λ alone]

$$\frac{\partial L}{\partial w} \Rightarrow w = \frac{x^T y}{(x^T x + \lambda I)}$$

$$\frac{\partial L}{\partial w} \Rightarrow w = (x^T x + \lambda I)^{-1} x^T y$$

Part 3 (RR using Gradient Descent)

$$L = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

Vector form loss fxn :

Regularizing.

$$L = (XW - Y)^T (XW - Y) + \lambda \|W\|^2$$

$$L = (XW - Y)^T (XW - Y) + \lambda W^T W$$

$$X = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1n} \\ 1 & x_{21} & x_{22} & \cdots & x_{2n} \\ 1 & \vdots & \vdots & \cdots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nm} \end{bmatrix}_{(n \times m)} \quad Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}_{(1 \times m)} \quad W = \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_n \end{bmatrix}_{(1 \times m)} \rightarrow n+1$$

$w_0, w_1, w_2, \ldots w_n$ (parameters)

$$w_0 = w_0 - \eta \frac{\partial L}{\partial w_0}, \quad w_1 - \eta \frac{\partial L}{\partial w_1}, \ldots \ldots, \quad w_n = w_n - \eta \frac{\partial L}{\partial w_n}$$

$$W_{new} = W_{old} - \eta \boxed{\frac{\Delta L}{\Delta W}} \rightarrow \text{gradient matrix} \begin{bmatrix} \frac{\partial L}{w_0} \\ \frac{\partial L}{\partial w_1} \\ \vdots \\ \frac{\partial L}{\partial w_n} \end{bmatrix}$$

$$L = (XW - Y)^T (XW - Y) + \lambda W^T W$$

$$L = (W^T X^T - Y^T)(XW - Y) + \lambda W^T W$$

Dividing by $\frac{1}{2}$ for mathematical convinience.

$$L = \frac{1}{2}(W^T X^T - Y^T)(XW - Y) + \frac{1}{2}\lambda W^T W$$

$$L = \frac{1}{2}\left[W^T X^T X W - W^T X^T Y - Y^T X W + Y^T Y\right] + \frac{1}{2}\lambda W^T W$$

$$L = \frac{1}{2}\left[\underline{W^T X^T X W} - \underline{2 W^T X^T Y} + Y^T Y\right] + \frac{1}{2}\lambda W^T W$$

$$\frac{\partial W^T X^T X W}{\partial W} \Rightarrow 2 X^T X W \quad , \quad 2 X^T Y \quad , \quad 2\lambda W$$

$$\frac{\partial L}{\partial W} = \frac{1}{2}\left[(2 X^T X W) - 2 X^T Y\right] + \lambda W$$

$$\frac{\partial L}{\partial W} = X^T X W - X^T Y + \lambda W$$

$$W = \begin{array}{cccc} w_0 & w_1 & \cdots & w_n \\ \end{array}$$
$$W = [0, 1, \ldots, 1]$$ starting point.

epochs.

$$W = W - \eta \frac{dL}{dW}$$

$W \rightarrow$ final answer.

Part 4

How the cofficients get affected ?

when $\lambda \uparrow$ what happens to cofficient

λ 0 to ∞

$\lambda = 0$: Linear Regression.

$\lambda \uparrow$: Cofficients shrinks/decrease towards 0, but never becomes zero.

Higher values are impacted more.

x_1 x_2 x_3 $\longrightarrow$ LR.

$w_1 = 1000$ $w_2 = 10$ $w_3 = 1$

After Ridge Reg. $\lambda = 1$ to ∞

Cofficient will decrease.

x_1 x_2 x_3 $\longrightarrow$ R.R

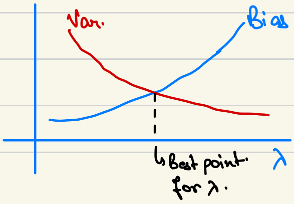
$w_?$ $w_?$ $w_?$

Higher Coefs. will decrease sharply than compare to coefs with lower values.

Bias Variance Trade-off.

$\lambda \downarrow$ close to 0 $\longrightarrow$ Bias $\downarrow$ and variance $\uparrow$.

$\lambda \uparrow$ $\longrightarrow$ Underfitting and variance $\downarrow$



Impact on the Loss Function

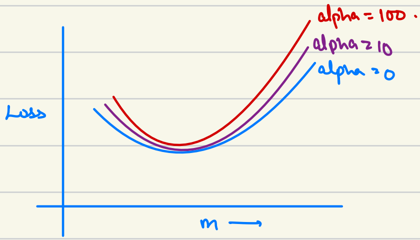
$$L = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \lambda \|w\|^2$$

$x, y \;=\; m, b \rightarrow b = 0$ (const.)

tunable

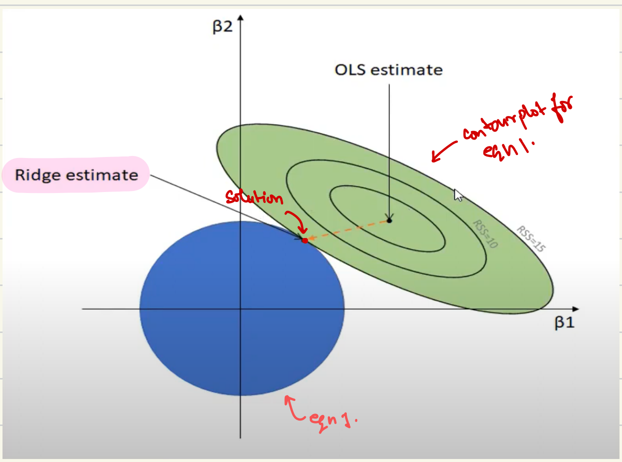
$$L = \sum_{i=1}^{n} (y_i - m x_i) + \lambda m^2$$

const



Loss fxn. moves towards center. (origin) as m↑

Why Ridge Regression is called so?



Since solution is at ridge (perimeter) of circle ∴ Ridge Regression.

Ridge Regression Tips:

If no. of input cols > 2

More $x_1, x_2, x_3 \ldots x_n$ → we have to regularize it.

Hard Constraint Ridge Regression

$$L = \underbrace{MSE}_{f x n 1} + \underbrace{\lambda \|w\|^2}_{f x n 2}$$

Let →

3 coefs = $\beta_0, \beta_1, \beta_2$

$(y_i - \hat{y})^2 \qquad \lambda m^2$

$\sum_{i=1}^{n} (y_i - (\beta_0 + \beta_1 x_{i_1} + \beta_2 x_{i_2}))^2$ — eqn 1.

$\lambda(\beta_1^2 + \beta_2^2)$ — eqn 2.

Lasso Regression

Also called L1 Norm.

In Ridge Regression : $L = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \lambda ||w||^2$

$\hookrightarrow \lambda (w_1^2 + w_2^2 + \dots + w_n^2)$

Lasso Regression : $L = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \lambda ||w||$

$\hookrightarrow \lambda (|w_1| + |w_2| + \dots + |w_n|)$

★★★ Note :—

$x_1 \quad x_2 \quad - - - \quad x_n$ ← High Dimension

Higher chances of overfitting in case of Regression.

In Lasso : As $\lambda \uparrow \longrightarrow 0$

So, the less important features are neglected. → dims decreases.

Basically, it's a kind of feature selection.

Important Questions :—

1. How are cofficients affected ?

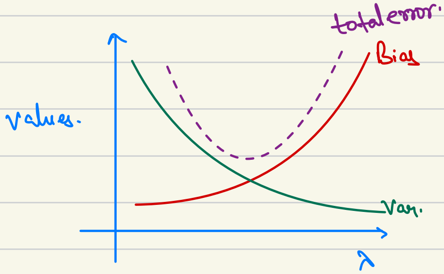
   Cofficient shrinks

   $\lambda \uparrow \longrightarrow$ coef $\simeq 0$ [Underfitting may occur]

2. Higher cofficients are affected more.

3. Impact of λ on Bias and Variance.

   $\lambda \uparrow$ Overfitting ↓ i.e Bias ↑ var ↓



4. Effects on Loss function.

Alpha is stuck at 0.

Very. Very. ★★★. imp.

Why Lasso Regression creates sparsity ?

Sparcity : Value becomes 0.

Single var. → x | y.

Simple Linear Regression : $y = mx + b$

$b = \bar{Y} - m\bar{X}$

$\bar{Y} \rightarrow mean(x)$

$\bar{X} \rightarrow mean(x)$

$$m = \frac{\sum_{i=1}^{n} (y_i - \bar{Y}) \cdot (x_i - \bar{x})}{\sum_{i=1}^{n} (x_i - \bar{X})^2}$$

For Ridge Regression : $m = \dfrac{\sum_{i=1}^{n} (y_i - \bar{Y}) \cdot (x_i - \bar{x})}{\sum_{i=1}^{n} (x_i - \bar{X})^2 + \lambda}$

For Lasso : $b = \bar{y} - m\bar{x}$

$$L = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \lambda |m|$$

$$L = \sum_{i=1}^{n} \left(y_i - mx_i - \bar{y} - m\bar{x}\right)^2 + \lambda |m|$$

We cant differentiate $|m|$ at point 0.

· when $m > 0$ $|m| = m$.

Mathematical Convinience.

$$\frac{\partial L}{\partial m} = \sum_{i=1}^{n} (y_i - mx_i + \underset{-\bar{y}}{m\bar{x}})^2 + 2\lambda |m|$$

$$\frac{\partial L}{\partial m} = 2 \sum_{i=1}^{n} (y_i - mx_i - \bar{y} + m\bar{x})(-x_i + \bar{x}) + 2\lambda = 0$$

$$\frac{\partial L}{\partial m} = -2 \sum_{i=1}^{n} \left[(y_i - \bar{y}) - m(x_i - \bar{x}) \right] (x_i - \bar{x}) + 2\lambda = 0$$

$$\frac{\partial L}{\partial m} = - \sum_{i=1}^{n} \left[(y_i - \bar{y})(x_i - \bar{x}) - m(x_i - \bar{x})^2 \right] + 2\lambda = 0$$

$$\frac{\partial L}{\partial m} = - \sum_{i=1}^{n} (y_i - \bar{y})(x_i - \bar{x}) + m \sum_{i=1}^{n} (x_i - \bar{x}) + \lambda = 0$$

$$\frac{\partial L}{\partial m} = m \sum_{i=1}^{n} (x_i - \bar{x})^2 = \sum_{i=1}^{n} (y_i - \bar{y})(x_i - \bar{x}) - \lambda$$

$$\Rightarrow \quad m = \frac{\sum_{i=1}^{n} (y_i - \bar{y})(x_i - \bar{x}) - \lambda}{\sum_{i=1}^{n} (x_i - \bar{x})^2}$$

Lasso :

$m > 0$ $m = \sum_{i=1}^{n} \frac{(y_i - \bar{y})(x_i - \bar{x}) - \lambda}{(x_i - \bar{x})^2}$	$m = 0$ $m = \frac{\sum (y_i - \bar{y})(x_i - \bar{x})}{\sum (x_i - \bar{x})^2}$
$m < 0$ $m = \frac{\sum_{i=1}^{n} (y_i - \bar{y})(x_i - \bar{x}) + \lambda}{\sum_{i=1}^{n} (x_i - \bar{x})^2}$	

$m > 0$

+ve i.e $m > 0$

$$m = \sum_{i=1}^{n} \frac{\boxed{(y_i - \bar{y})(x_i - \bar{x})} - \lambda}{(x_i - \bar{x})^2}$$

When $\lambda \uparrow$

Let $m = \frac{yx - \lambda}{x^2}$ $\left\{ \begin{matrix} yx = 100 \\ m = 50 \end{matrix} \right\}$

$m = \frac{100 - \lambda}{50}$

$\lambda = 0$	$\lambda = 10$	$\lambda = 50$	$\lambda = 100$
$m = 2$	$m = \frac{9}{5}$	$m = 1$	$m = 0$

$\xrightarrow{\hspace{2cm}}$ Shrinking of m.

As value increases above ($\lambda = 100$) than that of ($\bar{x}\bar{y} = 100$).
The formula changes to formula for $m < 0$

$$m = \frac{\bar{y}\bar{x} + \lambda}{\bar{x}^2}$$

$$= \frac{100 + \lambda}{50} \qquad \frac{\lambda = 150}{m = 5}$$

As this $m = 5$, the algo. stops at 0.

$m < 0$

$$m = \frac{\sum_{i=1}^{n} (y_i - \bar{y})(x_i - \bar{x}) + \lambda}{\sum_{i=1}^{n} (x_i - \bar{x})^2}$$

Since $\lambda > 0$ always for $m < 0$
therefore $\sum_{i=1}^{n} (y_i - \bar{y})(x_i - \bar{x}) \longrightarrow$ -ve. to make m '-ve'.

$$m = \frac{-100 + \lambda}{50}$$

$\lambda = 0$	$\lambda = 50$	$\lambda = 100$	$\lambda = 150$
$m = -2$	$m = -1$	$m = 0$	$m = 1$

At this point formula changes to $\frac{-100 - \lambda}{50}$

Now, $\lambda = 150$
$m = -5$

Situation worsens for m
$\therefore$ Algo. stops at 0.

Why no sparsity at Ridge Regression?

$$m = \frac{\sum (y_i - \bar{y})(x_i - \bar{x})}{\sum (x_i - \bar{x})^2 + \lambda}$$

$\lambda \uparrow$ $m \neq 0$ as Numerator is '+ve' not 0.

because $\frac{+ve}{\lambda}$ = always a term.

0 comes only when $\frac{0}{\lambda} \to 0$.

Because λ is at denominator.

ElasticNet Regression

It's a combination of Ridge Regression and Lasso Regression.

Ridge Regression.

$$L = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \lambda \|w\|^2$$

L^2 Norm

$\lambda \uparrow \quad m \downarrow \cong 0$

Lasso Regression.

$$L = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \lambda \|w\|$$

L1 Norm

$\lambda \uparrow \quad m \downarrow = 0$

ElasticNet → Combination of both.

$$L = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + a\|w\|^2 + b\|w\|$$

Hyperparameter → {λ, l1_ratio}

By default → $\lambda = 1$ i.e. $a = 0.5$, $b = 0.5$

l1_ratio = 0.5

if l1_ratio = 0.9

90% R.R 10% LR.

In scikitlearn :

$\lambda = a + b$

$l1\_ratio = \frac{a}{a+b}$

Also applied when inputs have multicollinearity.